Philips HealthSuite Digital Platform: Case Analysis, Cost Study and Exercise Plan

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I. Introduction

Medical data and imaging volumes keep growing, and this strains the IT that hospitals run on their own servers [3]. Philips answered with the HealthSuite Digital Platform (HSDP), a Platform-as-a-Service (PaaS) built on Amazon Web Services (AWS) that brings clinical records, medical images and data from connected devices into one regulated cloud environment [1], [2]. HSDP is a migration and transformation story. Philips first ran its own healthcare services on core AWS infrastructure and is now replacing them with AWS-managed services [4], [6].

This report has three parts. First, a case analysis (Sections II and III) covers the problems of on-premises healthcare IT, Philips' two-phase move, the outcomes it reports and the criticisms of it. Because Philips has not published the internals of HSDP, we build an assumed architecture in which every component is labelled as confirmed, confirmed in a prototype only, or assumed. Second, a cost analysis (Section IV) compares a managed AWS design with an on-premises design for the device-data pipeline at Philips' scale of 1.3 million connected devices [4]. Third, an exercise plan (Section V) describes Mini HealthSuite, in which simulated medical devices send vital signs to AWS in three stages: local, one virtual machine, and managed services. The stages mirror Philips' two phases.

The cost analysis gives two results. Before staff, the on-premises design costs less per month than the managed design, but with staff the answer depends on engineer counts that we could not source, so we report a break-even and not a winner. In the managed design, getting data in is the largest cost, so the reading interval and the ingestion route matter more than server size. Measured results of the exercise follow in a separate report. Section VI gives the team roles and timeline.

II. Case Study: Philips HSDP

A. Background and Timeline

Royal Philips is a Dutch health technology company that sells medical devices, imaging systems and clinical software to hospitals and consumers. In 2014 it launched the HealthSuite Digital Platform (HSDP) as a shared cloud foundation for its connected health products [5], [6]. Instead of each product team running its own servers and handling its own regulatory compliance, every product could build on one platform. Table I lists the main milestones.

Over these milestones, HSDP changed from a cloud for connected devices into a general Platform-as-a-Service for health data, imaging and AI. The device counts in 2015 (over 7 million) and 2025 (over 1.3 million) are not directly comparable: the 2015 figure counts devices and sensors together, while the 2025 figure refers to connected devices. We use the 2025 figure in our cost projections in Section IV because it is the most recent.

TABLE I
Key milestones of Philips HSDP on AWS

Year	Milestone	Ref.
2014	HSDP launched as Philips' cloud platform for connected health	[5], [6]
2015	Collaboration with AWS IoT; platform already managing over 7 million connected devices and sensors, aiming for hundreds of millions	[16]
2015	15 PB of patient data stored on the platform	[13]
2023	HealthSuite Imaging (cloud PACS) launched on AWS, using Amazon HealthLake Imaging (renamed AWS HealthImaging); first plans for generative AI on Amazon Bedrock	[17]
2024	Expanded collaboration on HealthSuite cloud services and generative AI workflows	[12]
2025	AWS named Philips' preferred cloud provider; over 100 AWS services in use; over 1.3 million connected devices	[4]

B. Problems with On-Premises Healthcare IT

Medical imaging shows how quickly on-premises healthcare IT comes under strain. About 3.6 billion imaging procedures are performed worldwide each year [3]. In the United States, the average radiologist's workload rose from 58 to 100 procedures a day between 2008 and 2023, and a single imaging study roughly doubled in size to about 150 MB [3]. Hospitals stored this data in picture archiving and communication systems (PACS) and Vendor Neutral Archives (VNA), and the growth overwhelmed them [3].

Running such systems on a hospital's own servers is costly and difficult for several reasons. First, hardware wears out and must be replaced. Data center operators surveyed by the Uptime Institute report refreshing servers about every 41 months, so a hospital faces a new round of purchases roughly every three and a half years [30]. Second, hospitals overprovision: they buy for peak load and future growth, so much of the capacity sits idle. A Philips and AWS prototype shows how large this effect can be, since giving every hospital device its own GPU would leave thousands of GPUs likely idle about 99% of the time [20]. Third, availability demands redundancy,

meaning extra copies of the data and a second site for disaster recovery, so much of the setup is paid for twice [3]. Fourth, servers need floor space, power and cooling whether or not they are busy. Fifth, people must install, patch, monitor and back up the systems, and skilled staff are hard to find: 53% of data center operators report difficulty finding qualified candidates [30]. Sixth, each system for images, reports and patient records is separate proprietary software with its own licence, which leaves data scattered across programs, a problem called data fragmentation [3]. Seventh, the hospital carries all of the compliance work itself, including HIPAA in the United States and GDPR in Europe [2].

Philips states that hospitals can reduce what they spend on their own hardware and data centres by moving their image-management platform to the cloud [17]. Section IV estimates these costs in numbers and compares them with AWS.

C. Migration in Two Phases

Philips' move to the cloud did not happen in one step. Following [4], we describe it in two phases. We interpret these phases using the same two terms as our exercise in Section V: *lift-and-shift* (moving onto cloud infrastructure while keeping your own software) and *refactoring* (replacing your own components with managed cloud services).

Phase 1: Moving onto AWS (from 2014). Philips first ran HSDP on core AWS building blocks: Amazon EC2 for compute [9] and Amazon S3 for storage [10]. At that time AWS offered no healthcare-specific managed services. Amazon HealthLake, its managed clinical data store, only became generally available in July 2021 [18], and AWS HealthImaging followed in 2023 [3]. Philips therefore built and operated its own services on top of AWS, including Container Host, notifications, a DICOM medical image store, a clinical data repository (CDR), a public key infrastructure (PKI) for certificates, and a Cloud Foundry application platform [5], [6]. In this phase Philips used AWS machines, but it still built and maintained most of the healthcare functionality itself.

Phase 2: Replacing its own services with AWS-managed ones. Once AWS launched healthcare services, Philips began retiring its own versions. Version 0.70.0 of Philips' official Terraform provider (October 2025) removed CDR and DICOM support because both services reached end-of-life, and version 1.3.0 (June 2026) decommissioned all resources for Container Host, Docker and PKI [6]. Philips now reports using more than 100 AWS services [4]. The shift is still in progress: Cloud Foundry is still listed as running on the HSDP status page [7]. Table II summarises these changes.

Hybrid extension: not everything moves to the cloud. Healthcare cannot move every system to the cloud. Safety-critical components must stay close to the patient, and some critical-care services must keep running even if a hospital loses its internet connection [19], [20]. Philips therefore uses hybrid designs in which on-premises and cloud components work together. It manages more than 70,000 on-premises servers across 1,200 locations, all of which need remote maintenance, monitoring and updates [19]. Since 2019 it has used AWS Outposts, which is AWS-managed hardware installed on site that runs the same AWS services and APIs as the cloud, so that on-premises and cloud workloads share the same development, deployment and management tools [19]. In its image-guided therapy prototype, a lightweight Linux gateway in the hospital sends images to AWS for processing while the imaging system itself stays on site [20]. The hybrid approach also helps with regulation. Medical devices need lengthy validation before every change, but cloud infrastructure that hosts only non-critical components is not classified as a medical device, so new features can be released without revalidating the whole system [20].

TABLE II
PHILIPS-BUILT HSDP SERVICES AND THEIR LIKELY AWS REPLACEMENTS

Philips service	Status [6]	AWS replacement
DICOM store	End-of-life (v0.70.0, Oct 2025)	AWS HealthImaging [3]
CDR	End-of-life (v0.70.0, Oct 2025)	Amazon HealthLake†
Container Host, Docker	Decommissioned from tooling (v1.3.0, Jun 2026)	Amazon EKS† [9]
PKI	Decommissioned from tooling (v1.3.0, Jun 2026)	AWS Private CA†

†Our assumption, based on the closest AWS managed service with the same role. Philips has not publicly named these replacements. EKS was used in a Philips prototype on AWS [9].

Why Phase 2 makes sense, and its trade-off. Managed services move routine work such as patching, scaling and compliance features from Philips to AWS, so Philips can spend its engineering effort on healthcare products instead of generic infrastructure. The cost is deeper dependence on a single provider, which we discuss as vendor lock-in in Section II-E.

Our exercise follows the same pattern on a small scale: Stage B (lift-and-shift to a VM) mirrors Phase 1, and Stage C (refactoring to managed services) mirrors Phase 2. It is also hybrid: the simulated medical devices run on a local machine that plays the role of the hospital and send their data to AWS.

D. Key Outcomes

Table III summarises the measurable results Philips and AWS report from moving HSDP workloads to the cloud.

Two of these results come from a prototype rather than the production platform: a near-real-time inference platform for image-guided therapy that returns results to the hospital within 5 seconds and cuts GPU cost 30-fold by sharing GPUs across devices and scaling them with demand [20]. The 67% figure is an expected saving from an ML tool that detects protected health information (PHI) in medical records, not a measured result [11]. Philips also reports a qualitative outcome: AWS services take over routine work so that its developers can focus on clinical workflows and faster product development [4].

TABLE III
REPORTED OUTCOMES OF PHILIPS WORKLOADS ON AWS

Metric	Value	Note	Ref.
Data managed on AWS	134 PB; target 1 EB by 2030	Includes nearly 11 billion medical images and records; likely covers customer imaging data, not only HSDP	[4]
Infrastructure cost	Up to 36% lower	Best case, achieved through data-centre migrations	[4]
Inference cost	30× lower	Prototype; compared with one cloud GPU per hospital device, not with on-premises	[20]
Image feedback latency	≤ 5 s end-to-end	Prototype; key-event detection during angioplasty	[20]
De-identification time	67% lower	Expected (projected), for HL7 data elements	[11]

Note: All figures are reported by Philips and AWS and have not been independently audited. None of the sources gives absolute costs, so Section IV builds its own cost estimates.

E. Criticisms

Vendor lock-in at two levels. HSDP is a proprietary platform. Its subscription terms allow Philips to change or remove platform features at any time and to reject client applications at its discretion, and they do not require Philips to return customer data when a contract ends [14]. Applications built on HSDP's APIs are therefore costly to move elsewhere. We observe that the same risk now applies one level down: by replacing its own services with AWS-specific ones such as AWS HealthImaging (Section II-C) and naming AWS its preferred cloud provider [4], Philips has made HSDP itself increasingly dependent on a single provider.

Dependence on network connectivity. Philips' patient-monitoring service terms require a continuous 24×7 connection to HealthSuite for certain services, and warn that a network outage can stop the central monitoring station from receiving real-time alarms from bedside monitors [15]. Philips' own prototype work confirms this from the engineering side: once AWS met the performance requirements, the only remaining limiting factor was the hospitals' last-mile network connection [20].

Limits of cloud and automation. As Section II-C showed, the cloud extends hospital systems rather than replacing them. Automation also has limits: Philips' ML tool for de-identifying medical records still misses about 3 in every 1,000 items of protected health information, so its output must be reviewed by humans [11].

Our exercise reflects the lock-in trade-off on a small scale: PostgreSQL on Amazon RDS can be moved to another provider with little change, whereas code written for AWS IoT Core is specific to AWS.

III. ASSUMED HSDP ARCHITECTURE

A. Approach

Philips has not published the internal architecture of HSDP, so we build a reference architecture from public sources. HSDP is organised into six domains: Connect, Authorize, Host, Store, Analyze and Share [8]. We use only these domains. Where a component runs, in the hospital or in AWS, is a separate question, so we show location in its own table column and in Figure 1 instead of inventing new domains. Every component carries one of three labels:

- *Confirmed:* a Philips or AWS source names it.
- *Confirmed (prototype):* it appears only in Philips' 2023 image-guided therapy prototype [20]. This shows how Philips builds on AWS but does not prove that all of HSDP works this way.
- *Assumed:* it is our inference, and the table gives the reason.

B. Services and Domains

Table IV lists the components we found or assumed, grouped by domain. Store and Share are grouped because Philips describes them together [2]. Twelve components are confirmed, three are confirmed only in the prototype, and four are assumed.

Only two HSDP components run in the hospital: the IoT Greengrass gateway (Connect) and AWS Outposts (Host). Everything else runs in AWS regions. The gateway sends device data to the cloud, and Outposts runs AWS services on site for critical-care solutions that must be available all the time [19]. Section III-C follows one piece of device data through these components.

C. Device-Data Flow

Figure 2(a) follows one piece of data through Philips' image-guided therapy prototype [20], the best-documented flow we found. The numbered steps match the arrows.

1) A lightweight Linux gateway in the hospital uploads the large image file, between 100 MB and several gigabytes, to Amazon S3. Philips reports that direct upload to S3 meets its performance requirements for files of this size.
2) The gateway then sends a small MQTT message through AWS IoT Core that says where the file is and how to process it, and the message is published to an Amazon SQS queue. The source does not say how IoT Core hands the message to the queue; we assume an IoT rule.
3) Workers on Amazon EKS watch the queue. For each task they fetch the algorithm's container from Amazon ECR and the image from S3 and run the algorithm. Machine-learning algorithms use NVIDIA Triton on GPU nodes.
4) The result returns through a results queue and an IoT topic to the hospital. Philips reports feedback within 5 seconds end to end. This latency covers image analysis,

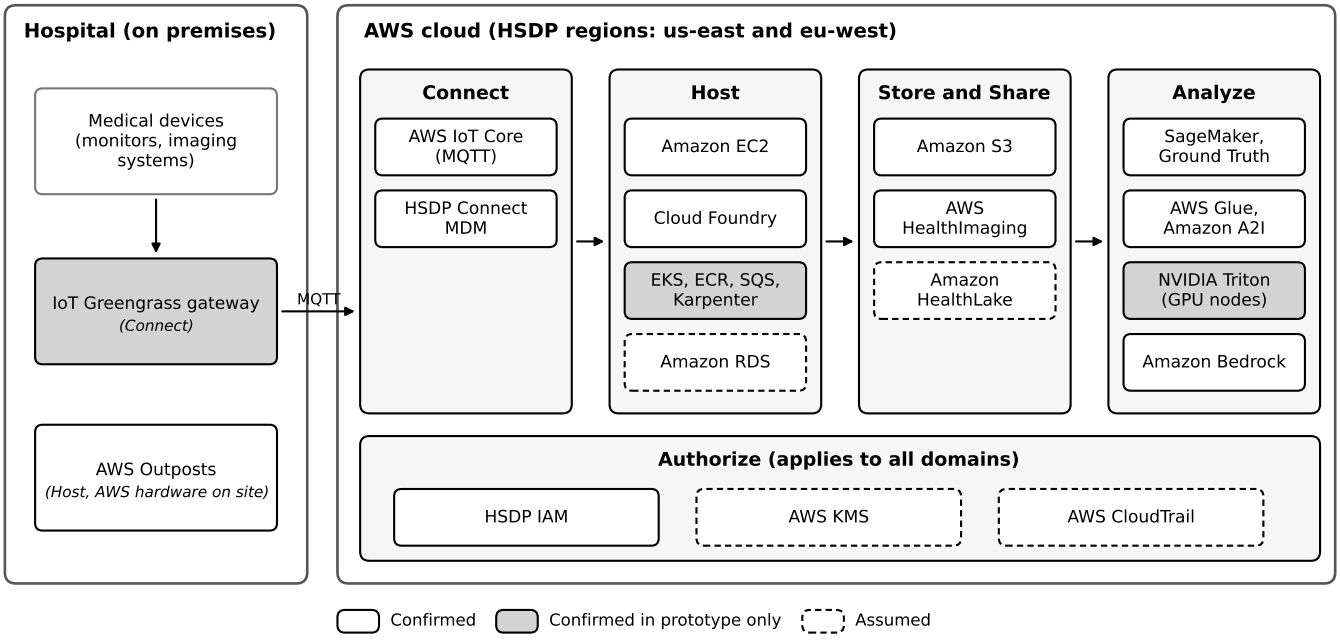


Fig. 1. Assumed HSDP architecture by domain and location. Solid outline: confirmed. Grey fill: confirmed in the prototype only. Dashed outline: assumed.

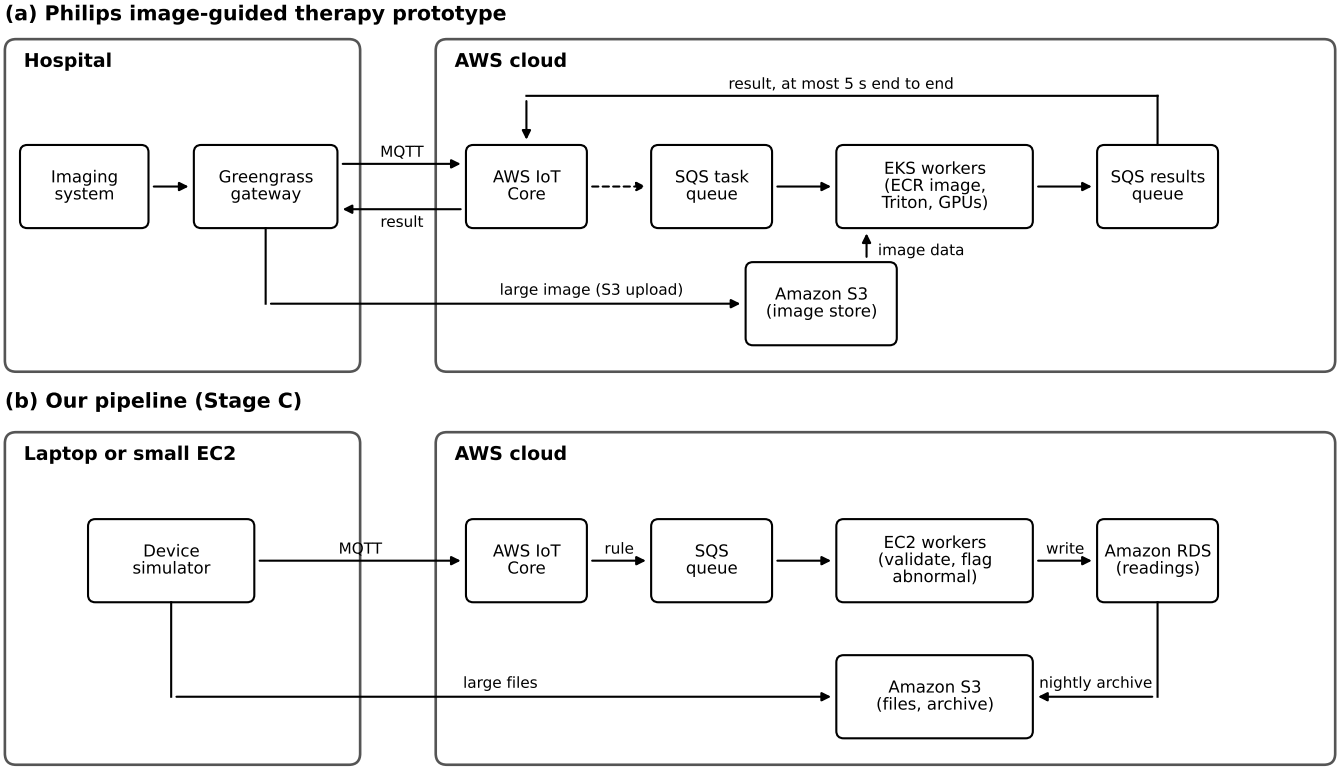


Fig. 2. Device-data flow. (a) Philips' image-guided therapy prototype [20]; the dashed arrow is our assumption (an IoT rule). (b) Our Mini HealthSuite pipeline in Stage C.

so it is not directly comparable with our vitals latency (Section V).

5) EKS nodes and pods are added or removed according to the length of the request queue, using a custom autoscaler and Karpenter.

Figure 2(b) shows our pipeline, which follows the same pattern at small scale. A simulated device publishes readings over MQTT to AWS IoT Core. A rule forwards each message

TABLE IV
HSDP COMPONENTS BY DOMAIN, WITH LOCATION AND CONFIDENCE

Domain	Component	Runs in	Status	Source or reason
Connect	AWS IoT Core (MQTT)	AWS	Confirmed	Device connection and messaging [9], [20]
	IoT Greengrass gateway (Linux device)	Hospital	Confirmed (prototype)	Forwards device data and sends MQTT messages [20]
	HSDP Connect MDM (device registry, firmware updates)	AWS	Confirmed	Provider documentation [26]
Authorize	HSDP IAM (Philips-built)	AWS	Confirmed	[2], [6]
	AWS KMS (encryption keys)	AWS	Assumed	Standard AWS service for encryption keys; HSDP stores health data that must be encrypted
	AWS CloudTrail (audit logs)	AWS	Assumed	Health data rules require audit trails; CloudTrail is AWS's audit log service
Host	Amazon EC2	AWS	Confirmed	[9]
	Cloud Foundry (application platform)	AWS	Confirmed	Listed as running on the HSDP status page [7]
	AWS Outposts (AWS hardware on site)	Hospital	Confirmed	[19]
	Amazon EKS, ECR, SQS, Karpenter	AWS	Confirmed (prototype)	Processing backbone of the prototype [20]
	Amazon RDS (managed databases)	AWS	Assumed	Host offers databases [2]; RDS is AWS's managed relational database
	Two regions: us-east and eu-west	AWS	Confirmed	[23]
Store and Share	Amazon S3	AWS	Confirmed	Data lake and image storage [10], [11], [20]
	AWS HealthImaging	AWS	Confirmed	[3], [17]
	Amazon HealthLake (clinical records)	AWS	Assumed	Likely successor to the retired CDR (Section II-C)
Analyze	Amazon SageMaker, SageMaker Ground Truth	AWS	Confirmed	[5], [10]
	AWS Glue, Amazon A2I	AWS	Confirmed	PHI de-identification workflow [11]
	NVIDIA Triton on GPU nodes	AWS	Confirmed (prototype)	Inference server in the prototype [20]
	Amazon Bedrock (generative AI)	AWS	Confirmed	Announced in 2023 and 2024 [12], [17]

to an SQS queue, EC2 workers validate each reading and flag abnormal values, and the readings are written to Amazon RDS. Large files go to S3 with an MQTT notification, as in panel (a), and a nightly job archives old readings from RDS to S3. We do not build the machine-learning step or the return path. We use EC2 workers instead of EKS because the EKS control plane alone costs about $0.10 per hour [29], roughly $73 per month, which is too much for our budget.

Three design patterns from this flow shape our work. First, data and events travel separately: large files go to S3, and a small MQTT message announces them. Second, a queue decouples ingestion from processing, so workers can scale with the queue length and readings wait safely if workers restart. Third, Philips shares GPUs across devices and scales them with demand, which it reports lowers cost by a factor of 30 compared with one GPU per device [20]. The same pay-for-use idea drives our cost model in Section IV.

IV. Cost Analysis of the MQTT Device-Data Pipeline

A. Scope and Scenarios

We estimate what it would cost to run, at Philips' scale, the device-data pipeline of our exercise (Section V), together with the security and monitoring services that a healthcare deployment needs: device ingestion over MQTT, a database for recent readings, an archive for older data, and the audit and monitoring services around them. We compare two ways of running this pipeline, each matching one stage of our exercise:

- **On-premises:** Philips-owned servers run the MQTT broker, application, database and storage (our Stage A, local).
- **Managed cloud:** AWS IoT Core, Amazon RDS and Amazon S3 replace the self-run components (our Stage C, refactored).

We do not cost a self-managed cloud option (the same software on virtual machines, our Stage B) at Philips scale. Stage B is built and measured at small scale in Section V, where it provides the lift-and-shift baseline.

We model the device fleet in two scenarios. The *simple scenario* treats all 1.3 million connected devices [4] as identical

vital-signs devices, so its numbers can be checked by hand. The *realistic scenario* splits the fleet into imaging systems, which produce large log files and medical images, and vital-signs devices, which send small readings. Comparing the two shows how much the device mix affects cost. The comparison of the two deployment options uses the realistic scenario only.

Cloud costs were estimated with the AWS Pricing Calculator using on-demand list prices for the us-east-1 region, as of November 2026. Each estimate was saved, and its share link is listed in the appendix. Nothing was deployed at this scale; our own small-scale measurements (Section V) are used only to calibrate per-message cost and worker capacity.

Both options include the cost of the staff who run them. Operating on-premises servers needs IT staff for patching, monitoring and hardware replacement, and managed services reduce this work without removing it. We estimate the number of engineers each option needs and price them with published salary data (Section IV-C). We assume four generic engineers for the on-premises option and two for the managed option, and price both with the same median wage for network and computer systems administrators [36]. Neither count is sourced. Google's guidance is eight engineers for a round-the-clock rotation with a primary and a secondary on call, and six per site for a two-site team [37], so the sensitivity analysis varies the on-premises count from two to twelve.

B. Assumptions

Tables V–VII list the inputs to the cost model. Values marked (A) are our assumptions and values marked (M) will be measured in our exercise (Section V); all other values come from the cited source.

TABLE V
TRAFFIC ASSUMPTIONS

Assumption	Simple	Realistic
Connected devices	1.3M [4]	1.3M [4]
Imaging systems	–	16,000 [21]
Vital-sign devices	1.3M	1.284M
Reading interval	1 min (A)	5 min (A)
Vital-sign message size	0.5 KB (M)	0.5 KB (M)
Log messages per imaging system per day	–	300,000 [21]
Log transport	–	Daily file + MQTT notification (A)
Log message size	–	0.2 KB (A)
Studies per imaging system per day	–	15 [22]
Study size	–	150 MB [3]
Systems uploading images	–	10% (A)
Connection time	24×7 [15]	24×7 [15]

The imaging system count is a 2015 figure and is likely higher today [21]. The log rate of 300,000 messages per day is the average for an MRI scanner [21]; we apply it to all imaging modalities. Philips analyses these logs daily [21], so we assume each system uploads one daily log file to Amazon S3 followed by a small MQTT notification, the same pattern Philips used for images in its prototype [20]. The 15 studies per day come from one hospital's example [22]. We assume all devices connect over MQTT, as in Philips' image-guided therapy prototype [20]; Philips has not published which protocols its whole fleet uses.

TABLE VI
PEAK-TO-AVERAGE LOAD BY TRAFFIC TYPE

Traffic	Peak/avg	Reasoning
Vital signs	≈1× (A)	Readings arrive at fixed intervals, 24×7
Images	2.4× (A)	15 studies spread over about 10 working hours instead of 24
Log files	3× (A)	Uploaded during an 8-hour night window
Stress test	4×	Pareto (80/20) pattern; used only in the sensitivity analysis

Peak load matters only for components we size in advance: the on-premises servers and, in the managed option, the EC2 workers and the database instance. AWS IoT Core charges per message and scales automatically, so peaks do not raise its cost. Vital signs can still burst above the average when many devices reconnect at once, for example after a network outage; the 4× stress test covers this case.

TABLE VII
STORAGE AND PRICING RULES

Rule	Value
Database	Last 30 days of readings (A)
S3 Standard	Data up to 90 days old (A)
Older data	S3 Glacier Deep Archive
Storage priced at	Volume stored at month 12 (A)
Prices	us-east-1, on-demand, AWS Pricing Calculator, November 2026

The tiering follows AWS's best practice for IoT systems: keep only recent data in short-term storage and move older data to cheaper storage classes through lifecycle policies [24]. AWS applies the same hot, warm and cold pattern in its own IoT SiteWise service [25]. The 30-day and 90-day periods are our own choice. We price storage at month 12 because the AWS Pricing Calculator prices a fixed monthly volume and cannot model data that grows month by month.

Two inputs are derived rather than assumed. The number of on-premises servers follows the same sizing as the managed option: worker capacity comes from our measurements, and the number of MQTT brokers comes from a published limit on connections per broker (A), because every device stays connected 24×7. The on-premises cost inputs (hardware, power and cooling, staff, storage and backup copies) are sourced in Section IV-C.

Table VIII shows the monthly data volumes these assumptions produce.

These assumptions have three limitations. First, the imaging system count dates from 2015 while the total device count

TABLE VIII
MONTHLY DATA VOLUMES IMPLIED BY THE ASSUMPTIONS

Data stream	Simple	Realistic
Vital-sign messages (MQTT)	56.2 billion (28 TB)	11.1 billion (5.5 TB)
Imaging log files (S3)	–	480,000 files (28.8 TB)
Medical images (S3)	–	720,000 studies (108 TB)
MQTT notifications for files	–	1.2 million

dates from 2025, because no newer public breakdown exists. Second, the log rate is an MRI average applied to all imaging modalities. Third, list prices ignore the volume discounts Philips almost certainly receives, so our cloud costs are upper estimates.

C. Monthly Cost of the Two Options

This subsection prices the pipeline of Section IV-A in the realistic scenario. The managed cloud option is priced with the AWS Pricing Calculator; the links to the saved estimates are in the appendix. The on-premises option is priced from the inputs in Table XI. Both options use the same six cost components, so each row of Table IX compares like with like.

TABLE IX
MONTHLY COST, REALISTIC SCENARIO (USD)

Cost component	On-prem 1 site + backup	On-prem 2 active sites	Managed cloud
Device ingestion and queue	2,676.15	2,676.15	21,676.31
Application servers	1,338.08	2,676.15	379.36
Database	2,676.15	2,676.15	1,798.26
Archive storage	5,624.56	5,624.56	10,658.74
Security and monitoring	0.00	0.00	34.43
Network and facilities	1,023.98	2,050.80	included
Total without staff	13,338.93	15,703.83	34,547.10
Staff (4 and 2 engineers)	33,043.33	33,043.33	16,521.67
Total with staff	46,382.26	48,747.16	51,068.77

Totals may differ from the sum of the rows by up to 0.02 because of rounding. Managed prices include network, facilities, backup and availability. "1 site + backup" includes a backup site with the standby database, one archive copy, a tape drive and a switch. On-premises security and monitoring is covered by staff.

In the managed column, device ingestion and queue is AWS IoT Core plus Amazon SQS, application servers is Amazon EC2, the database is Amazon RDS, archive storage is Amazon S3 with S3 Glacier Deep Archive, and security and monitoring is AWS CloudTrail plus Amazon CloudWatch. The managed estimate assumes a Multi-AZ database, that is, a standby copy in a second data centre, because health data must stay available. Our exercise uses a single-AZ database to stay within budget. The annual cost of the managed option without staff is about $414,600.

On-premises estimate. Philips would have to buy and run the same pipeline on its own servers. Table X gives the server counts, which follow our sizing for the managed option. We convert purchases into a monthly cost by spreading the hardware price over its useful life and adding power and cooling, storage with backup copies, and staff.

We priced the on-premises option in two designs. The first has one active site plus a backup site that holds the standby database, one archive copy, a tape drive and a switch. It uses five Dell PowerEdge R760 servers [31]: two MQTT brokers, one application host and two database servers. A vendor benchmark reports about five million connections per broker node [34], so two brokers cover our 1.3 million devices, including one spare. The second design follows the managed option's standby in a second data centre: both sites are active and can carry the full load, with one broker and one application host per site and the database primary and standby split across sites, six servers in total, plus a 1 Gbps link between the sites. The link carries about 0.44 Gbps on average if one remote copy of all new data is replicated, and replication can queue at peaks. All servers use the same configuration for simplicity, which oversizes the application and database hosts. Each server's price is spread over 41 months, the refresh cycle operators report on average [30]. Power uses a measured server power curve [32], a PUE of 1.52 [30] and the US commercial electricity price [33]. The archive keeps three copies, following the three-copy and off-site parts of the 3-2-1 backup rule [38]: the newest 90 days on enterprise disks in two disk arrays, and older data on LTO-9 tape with two tape drives. The disk arrays are recertified units with a 30-day warranty and each takes at least 28 drives, so the backup site is raised to 28 drives; the other hardware is priced new. Each part of the data sits on one medium, so the rule's two-media part is not met. Table XXIV lists every hardware price, its source and the date opened, and Table XXV shows the calculation step by step.

Before staff, the on-premises option costs far less per month than the managed option, but it needs about $505,000 of purchases up front (about $556,000 for two active sites). The managed prices already include the provider's backup, high availability and network capacity, which the on-premises column only partly matches. Neither engineer count is sourced, so the robust result is the break-even: the monthly gap before staff is $21,208.17 (one active site) or $18,843.27 (two active sites), and an engineer costs $8,260.83 per month, so the on-premises option is cheaper only if it needs fewer than about 2.6 or 2.3 more engineers than the managed option. If both options need the same on-call rotation, the on-premises option is cheaper by that gap. If the managed option needs only two engineers while the on-premises option follows Google's counts (eight, or twelve for two sites), the managed option is cheaper by a wide margin (Table XVI). Varying the hardware life from 36 to 72 months does not change these conclusions. Support contracts, cabling, bandwidth beyond the inter-site link, the power of the disk arrays and tape drives, minimum charges at a second site, and a commercial licence for the MQTT broker [35] are not included, so the on-premises figures

are a partial estimate. The benchmark page does not state which EMQX edition it ran; if it ran a commercial edition, excluding the licence favours the on-premises option.

TABLE X
ON-PREMISES SIZING, REALISTIC SCENARIO

Component	1 site + backup	2 active sites
Workers (VMs)	6 (2 vCPU, 4 GiB each)	6; each site's host can run all
Web servers (VMs)	2	2; each site's host can run both
Database	1 primary + 1 standby at the backup site	1 primary at one site, 1 standby at the other
MQTT brokers	2 (1 + 1 spare)	1 per site
Application hosts	1	2
Physical servers	5	6
Archive storage	1,702 TB before copies; 3 copies (2 + 1 across sites)	same
Disks and tape	64 disks of 24 TB (28 per array at least), 213 LTO-9 cartridges	same

Workers, web servers and the database are the same workload as the managed option (database: 8 vCPU, 64 GiB, 1.75 TB SSD each); on-premises they run as virtual machines on the physical servers, all of one configuration. Brokers: 1.3 million connections divided by about 5 million per broker, rounded up, plus one spare. Disks hold the newest 90 days, tape the older data, all copies counted. The two disk arrays are recertified Dell ME5084 units, each with two SAS controllers; each array takes at least 28 drives.

TABLE XI
ON-PREMISES COST INPUTS

Input	Value	Source
Price per server	$51,822.96	[31]
Hardware life	41 months	[30]
Power and cooling per server per month	$74.10	[30], [32], [33]
Storage cost per TB, disk / tape	$91.63 / $6.08	Tab. XXIV
Backup copies kept	3	[38]
Storage hosting (2 disk arrays, 2 tape drives)	$65,781 (purchase)	Tab. XXIV
Network and space per month, 1 site + backup	$1,023.98	Tab. XXIV
Inter-site link per month	$928.95	Tab. XXIV
Engineers, on-premises / managed	4 / 2 (A)	[37]
Salary per engineer per year	$99,130	[36]

(A) = our assumption. Storage cost per TB is the raw media price.

Managed cost by service. Table XII breaks the managed cost down by AWS service for both scenarios.

Ingestion dominates the managed bill. IoT Core and SQS together make up 62.7% of the realistic scenario and 92.0% of the simple scenario, while the servers that process the data make up about 1%. In the realistic scenario ingestion costs about $1.95 per million messages; our exercise measures the same quantity at small scale (Section V). Archive storage is the second-largest cost at 30.9%, because medical images make up 76% of the stored data. Moving from the simple to the realistic fleet reduces the monthly cost by a factor of 2.8, mainly because readings arrive every five minutes instead of every minute, which cuts ingestion costs by about three quarters.

TABLE XII
MANAGED CLOUD COST BY AWS SERVICE (USD PER MONTH)

AWS service	Simple	Realistic
IoT Core	61,415.20	16,350.71
SQS	26,956.80	5,325.60
RDS (Multi-AZ)	4,815.52	1,798.26
S3 and Glacier Deep Archive	1,533.21	10,658.74
EC2, CloudWatch, CloudTrail	1,315.79	413.79
Total	96,036.52	34,547.10

Routing the readings. How the readings travel from AWS IoT Core to our application changes the ingestion cost considerably. Table XIII compares three routes for the realistic scenario.

TABLE XIII
INGESTION ROUTES, REALISTIC SCENARIO (IOT CORE PLUS SQS)

Route	How readings reach the application	USD per month
1	Application subscribes directly to IoT Core	20,809.91 (calculator)
2	IoT rule forwards each message to an SQS queue, workers read the queue	21,676.31 (calculator)
3	Basic Ingest: devices publish straight to the rule, skipping the broker	13,209.46 (calculator)

Route 2 is the pattern in Philips' prototype, where messages published to IoT Core are forwarded to an SQS queue and read by workers [20], so it is our main design and the one priced in Table IX. In Route 1 every message is billed twice, once when the device publishes it and once when IoT Core delivers it to the application [27]. Route 1 is priced as a separate estimate (IoT Core only, no queue). Route 2 adds a rule and a queue to that cost, about 4% more, but the queue also acts as a buffer: readings wait safely while workers restart, which makes it easier to show that no readings are lost during our migration test. Route 3 uses AWS IoT Basic Ingest, so no message is billed [27]. It costs 39% less than Route 2 and about 37% less than Route 1, but other services, such as a live dashboard, can no longer subscribe to the readings.

Section IV-D shows how storage tiering reduces the archive line, and Section IV-E tests how the totals change when our assumptions change.

D. Storage Tiering Savings

Archive storage is the second-largest cost of the managed option, and medical images make up 76% of the stored data (Section IV-B). Amazon S3 offers storage classes that trade

price against retrieval speed, and lifecycle rules move data from one class to another automatically. Following AWS's best practice for IoT data [24], we compare four policies for the data stored at month 12 in the realistic scenario. In every policy, data stays in S3 Standard for its first 90 days. Table XIV shows the results from the AWS Pricing Calculator; the saved estimates are listed in the appendix.

TABLE XIV
STORAGE POLICIES, REALISTIC SCENARIO

Policy	Data older than 90 days goes to	Retrieval of old data	USD per month	Saving vs A
A	Nothing moves; stays in S3 Standard	Immediate	36,314	–
B	S3 Glacier Deep Archive	Hours	10,659	71%
C	S3 Glacier Instant Retrieval	Immediate	14,475	60%
D	Images: Instant Retrieval; readings and logs: Deep Archive	Images immediate, others hours	13,562	63%

Moving old data to a cheaper class cuts the archive bill by 60 to 71%. The cheapest option is not automatically the best one, however. Deep Archive costs about a quarter of Instant Retrieval per gigabyte, but getting data back takes hours [28]. We assume clinicians sometimes need a patient's earlier images quickly, for example to compare them with a new scan, so slow retrieval of old images may not be acceptable. Policy C keeps everything instantly available and costs about $3,800 per month more than Policy B.

Policy D matches the storage class to the data. Old readings and logs are rarely read again, so they go to Deep Archive. Old images stay instantly available in Instant Retrieval. It costs about $2,900 more than Policy B and about $900 less than Policy C, while keeping fast access to the 76% of the data that is most likely to be needed. We therefore recommend Policy D if old images must stay quickly available, and Policy B if they do not.

The comparison ignores the per-gigabyte fees for reading data back out of the archive classes. These fees would raise the cost of Policies B, C and D if old data is read often. In summary, lifecycle tiering cuts storage cost by 60 to 71% for the same data, and how far to go depends on how quickly old images must be retrievable.

E. Sensitivity Analysis

The totals above depend on assumptions that we cannot verify, so we change one assumption at a time and record the new monthly total from the AWS Pricing Calculator. The baseline is the realistic scenario with the managed option and Route 2. Each case is a saved estimate; the links are in the appendix.

Each case changes only the parts of the pipeline that it affects. Case 1 changes the number of readings, so it affects ingestion, the database, the archive and the workers. Cases 2a and 2b change only the volume of images stored in Amazon S3. Case 3 changes only IoT Core and SQS: a batch of 100 events is a 20 KB message, which AWS bills as four 5 KB units. Case 4 changes only the servers we size ourselves, the workers and the database instance; IoT Core, SQS and S3 charge for use and do not change with peak load.

TABLE XV
SENSITIVITY OF THE MANAGED OPTION, REALISTIC SCENARIO (USD PER MONTH)

Case	Change from the baseline	Total	Change
Base	Realistic scenario, Route 2	34,547.10	–
1	Vital signs sent every 1 minute instead of every 5	105,319.99	+204.9%
2a	5% of imaging systems upload images (baseline 10%)	30,481.39	−11.8%
2b	50% of imaging systems upload images	65,867.89	+90.7%
3	Imaging logs streamed over MQTT in batches of 100 events instead of daily files	40,998.64	+18.7%
4	Peak load of $4\times$ (Pareto stress test), workers and database sized for it	36,651.74	+6.1%

The total is most sensitive to the reading interval. Sending vital signs every minute instead of every five minutes roughly triples the monthly cost (+204.9%), because ingestion is billed per message. The share of imaging systems that upload images comes second. Storage cost follows the amount of image data: halving the share saves 11.8%, while multiplying it by five adds 90.7%. Streaming logs over MQTT adds 18.7%, and a $4\times$ peak adds only 6.1%, because only the servers we size ourselves change with peak load. The two assumptions with the largest effect, the reading interval and the image share, are both our own assumptions, so we report the managed cost as a range and not as a single figure.

Each case keeps all other assumptions at their baseline values, so combinations of changes are not covered.

Table XVI varies the number of on-premises engineers.

TABLE XVI
MONTHLY TOTAL WITH STAFF (USD) BY NUMBER OF ON-PREMISES ENGINEERS; MANAGED CLOUD FIXED AT 2 ENGINEERS (51,068.77)

Engineers	1 site + backup	2 active sites	Cheaper option
2	29,860.59	32,225.49	On-premises
4	46,382.26	48,747.16	On-premises
6	62,903.93	65,268.83	Managed cloud
8	79,425.59	81,790.49	Managed cloud
12	112,468.93	114,833.83	Managed cloud

The managed count of two is also unsourced. At equal headcount the on-premises option is cheaper by 21,208.17 (1 site + backup) or 18,843.27 (2 active sites).

F. Discussion: On-Premises vs Cloud

Where the money goes. In the managed option, getting data in costs more than storing or processing it. Ingestion is 62.7% of the realistic bill, archive storage is 30.9%, and the servers that process the data are about 1% (Table XII). Because servers are such a small share, the savings come from two design choices and not from server sizing: routing readings through Basic Ingest cuts ingestion cost by 39% (Table XIII), and lifecycle tiering cuts storage cost by 60 to 71% (Table XIV).

On-premises against cloud. Without staff, the on-premises option costs 13,338.93 per month with one active site and 15,703.83 with two active sites, against 34,547.10 for the managed option, so it is cheaper by 21,208.17 or 18,843.27. Staff narrows this gap. With staff, the on-premises figures are 46,382.26 and 48,747.16 against 51,068.77 for the managed option, a difference of 4,686.51 or 2,321.61 (Table IX), because we assume four engineers on-premises and two for the managed option. Philips reports infrastructure cost reductions of up to 36% after data-centre migrations [4]. Our result, that the on-premises option is about 9% or 5% cheaper with staff, goes in the opposite direction and is not directly comparable, because Philips' figure covers its whole infrastructure and ours covers one device-data pipeline.

Effort and risk. The cost figures leave out part of what managed services provide. They take over patching, backups and scaling, which is why the staff line differs between the options. The two options also do not give the same service level. The managed prices include the provider's standby copy in a second data centre, managed backups and network capacity, and access to further managed services, such as machine learning and analytics, that an on-premises build would have to set up and run itself or do without. Our comparison covers ingestion and storage only, with no machine learning or analytics in either column, and it uses generic engineer counts, so it does not price specialised staff. The on-premises figures are therefore lower partly because they buy less. In exchange, the pipeline depends on AWS-specific services, the lock-in risk discussed in Section II-E: code written for IoT Core and SQS must be rewritten to move to another provider, whereas PostgreSQL on Amazon RDS is portable.

Limits of the estimate. These results rest on list prices and on the assumptions in Section IV-B, especially the reading interval and the share of imaging systems that upload images (Section IV-E). We therefore read the totals as orders of magnitude, not as forecasts. On these assumptions the on-premises option is cheaper before staff, and with staff it stays cheaper only if it needs fewer than about 2.6 more engineers than the managed option (2.3 for two active sites), so the result depends on a headcount that we could not source.

V. Exercise Plan: Mini HealthSuite

The exercise follows the steps set by the course: build a two-tier application locally, lift and shift it to a cloud virtual machine (VM), then refactor it to managed services and compare effort and cost. We make it a small version of the Philips device-data path described in Section III-C. Simulated medical devices send vital signs to the cloud, and we measure what happens at each stage.

A. Link to the Case Study

Every part that we build copies something Philips did and produces a number that we measure (Table XVII). Parts marked *optional* are built only if time permits.

TABLE XVII
What we build and how it links to Philips

We build	Philips basis	We measure
Devices send readings over MQTT to AWS IoT Core	IoT Core and MQTT in HSDP and in the prototype [9], [20]	Delay, lost readings, cost per million messages
A queue (SQS) between IoT Core and the workers	Prototype: IoT Core, queue, workers [20]	Readings lost when workers restart
The database moves from the VM to Amazon RDS	Philips' two migration phases (Section II-C)	Downtime, lost data, effort, cost
A simulator outside AWS acts as the hospital	Hospital gateway sending data to the cloud [20]	Delay from outside AWS to the cloud
Optional: large files go to S3, then a small MQTT message	The prototype uploads images to S3 and then sends a small MQTT message [20]	Upload time for 10, 50 and 100 MB
Optional: archive in S3 with patient names removed	S3 data lake [10] and PHI de-identification [11]	Storage cost

B. Application Design

The application has two ways to receive data: a web form for entering one reading, which is the basic two-tier application, and device input over MQTT. A plain HTTP endpoint stays as a baseline for comparison. Both entry points call the same rules, so the HTTP and MQTT results differ only in how the data travels. The code is split into small parts: settings, database access, reading rules, web pages and the MQTT input. Moving between stages therefore mostly means changing the settings.

Table XVIII lists the fields of one reading. Patient names are kept in a separate table, so any data that we pass on or archive can leave them out. A reading outside a sensible range (heart rate 20 to 250 beats per minute, SpO_2 50 to 100%, temperature 30 to 45 °C) is rejected. A valid reading outside normal limits, which are kept in the settings, is flagged as abnormal.

C. Architecture Across Stages

We build the exercise in three stages that match the two phases of the Philips migration (Section II-C). Table XIX shows what each stage uses.

Moving from Stage A to Stage B changes nothing: the same containers are copied to the VM. Moving from Stage B to Stage C changes the database address in the settings. IoT Core also needs device certificates and permission policies, and the

TABLE XVIII
FIELDS OF ONE READING

Field	Purpose
`device_id`	Which device sent the reading
`patient_ref`	A code for the patient, never a name
`heart_rate`	Beats per minute
`spo2`	Blood oxygen, in percent
`temperature`	Degrees Celsius
`seq`	Counter per device; a gap shows a lost reading
`sent_at`	Time the device sent the reading
`received_at`	Time we stored it; the difference is the delay
`abnormal`	True if a value is outside normal limits

TABLE XIX
THE THREE STAGES

	A: local	B: one VM	C: managed
Runs on	Laptop, Docker Compose	One EC2 VM, same Compose file	EC2 plus managed services
Database	PostgreSQL in Docker	PostgreSQL on the VM	Amazon RDS
MQTT broker	Mosquitto in Docker	Mosquitto on the VM	AWS IoT Core
Readings reach the workers	Straight from the broker	Straight from the broker	IoT rule, SQS queue, workers

application needs a new input part that reads from the queue. If time runs short, the application can subscribe to IoT Core directly instead of using the queue. We measure the effort of each step in hours. RDS runs in a single zone to save money, although the cost model in Section IV assumes a standby copy. We keep the exercise within our available AWS credits and use budget alerts to stop overspending.

D. Experiment Plan

We run one required test and two optional ones. The required test sends vital signs, many small steady messages, once over HTTP and once over MQTT, to each stage. The optional tests send large files (10, 50 and 100 MB to S3 followed by an MQTT message) and imaging logs streamed over MQTT. Table XX gives the fixed settings, so that every stage is tested the same way.

We measure the average delay and the delay of the slowest 5% of readings, lost readings (gaps in `seq`), CPU use, migration downtime and lost data, cost per million messages from the AWS bill, and effort in hours and steps. We use Locust to send vital-sign readings over HTTP and a short Python script with the paho-mqtt library to send them over MQTT, each reading being a 0.5 KB JSON message. The load rises in four steps of 10, 50, 100 and 500 messages per second, two minutes per step, repeated three times, and a short restart test stops the worker for 20 seconds at 100 messages per second to count lost readings. The load generator runs on a second small EC2 instance in the same region as the system under test, except in the local stage, where it runs on the same laptop. Delay is the time between the generator sending a reading and the application receiving it, with both EC2 machines synchronised to Amazon Time Sync, and the scripts and settings are in the `loadtest` folder of our GitHub repository. Before we run the tests, we expect the following:

1) RDS adds a few milliseconds per write, because the database is on a separate machine.
2) The queue adds a little delay but prevents lost readings when the workers restart.
3) The database on the VM reaches its limit first.
4) At our small size RDS costs more than a database on the VM, but it needs less ongoing work, such as backups and updates.

TABLE XX
FIXED TEST SETTINGS

Setting	Value
Message size	0.5 KB
Load steps	10, 50, 100 and 500 messages per second
Time per step	2 minutes
Messages per run	At most 100,000, to control cost
Load generator	A second small EC2 VM in the same region, so that home internet delay is not counted
Repeats	3

E. Optional Extensions

If time permits, we add four extras, each linked to the Authorize domain (Section III): an IAM role instead of stored access keys, AWS Systems Manager Parameter Store for the database password, an AWS CloudTrail audit log, and an S3 archive with patient names removed. The final report will state which of them were built. The cost model in Section IV prices CloudTrail and CloudWatch because it models a production deployment. Failure and recovery testing and data residency, which means keeping each region's patient data in its own region, are left as future work.

VI. TEAM ROLES AND TIMELINE

Jeet Daiya leads the build and the cost model. The other two members take well-defined tasks, so that each can explain his or her part in the viva. Table XXI shows who owns what.

“‘ “‘

TABLE XXI
ROLES OF THE TEAM MEMBERS

Member	Report sections
Jeet Daiya	III, IV (except the on-premises costing), V
Kavya Parmar	III, IV-C (on-premises), V-D, price and calculation tables in the appendix
Bhavya Thakkar	II, III, reference list

TABLE XXII
TIMELINE

Week	Work	Output
2	Case analysis, architecture, cost model, exercise plan and roles	This report
3	Build the three stages, run the load tests, and collect delay, lost readings, scaling behaviour and cost	Measured results, in a separate report
4	Finish and submit the report	Final report
5	Viva preparation	Viva

Table XXII gives the timeline that follows the course plan.

Every member reads the whole report before each submission, and Jeet checks that each section matches what was built. The scripts, settings and code are in the project's GitHub repository, which shows who did what.

APPENDIX A
AWS PRICING CALCULATOR ESTIMATES

Table XXIII lists every saved AWS Pricing Calculator estimate used in Section IV.

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TABLE XXIII
SAVED AWS PRICING CALCULATOR ESTIMATES

Estimate	Description	Monthly total (USD)	Link
1	Realistic, managed, Route 2	34,547.10	https://calculator.aws/#/estimate?id=adb9984493f4a9fc66e93f34de719cf07c79043c
2	Simple, managed, Route 2	96,036.52	https://calculator.aws/#/estimate?id=ce40738d685559a41cfab993655b79120b58ed45
3	Realistic, Route 3 (Basic Ingest)	13,209.46	https://calculator.aws/#/estimate?id=1553ba2b655ea3193dc704671ac0daf3c19d5250
4A	Storage policy A	36,313.78	https://calculator.aws/#/estimate?id=c9f633e6dd7352db0c9695563bf1b13e93aa2591
4B	Storage policy B	10,658.74	https://calculator.aws/#/estimate?id=1e69d6e44c5a6def87247c66e5d9e6ca947c91ef
4C	Storage policy C	14,475	https://calculator.aws/#/estimate?id=c0ad79970b5ff31b1d34ebfbdafac69ccf9c7289
4D	Storage policy D	13,561.58	https://calculator.aws/#/estimate?id=8f8fa0fb69c68a542b4d0ca30fead46393611db0
Case 1	Vital signs every 1 minute	105,319.99	https://calculator.aws/#/estimate?id=e437175efe59cbee799788a40508856e8586e410
Case 2a	5% image share	30,481.39	https://calculator.aws/#/estimate?id=4ae42224790cb3f5ca858d4685c50c9547a339e3
Case 2b	50% image share	65,867.89	https://calculator.aws/#/estimate?id=9d488ca6c2b8ea0f7b862ca47a70b81358e63941
Case 3	Imaging logs streamed over MQTT	40,998.64	https://calculator.aws/#/estimate?id=87a935c404c099b424cba3ec97a1123a7eab4b05
Case 4	$4\times$ peak load	36,651.74	https://calculator.aws/#/estimate?id=e044ad3d2e48855a2373151e746396df71bd563c
Route 1	IoT Core only, application subscribes directly	20,809.91	https://calculator.aws/#/estimate?id=0569f93009c9846b4ab4b7dd5b6469161eada454

TABLE XXIV
PRICES AND PARAMETERS USED FOR THE ON-PREMISES ESTIMATE, WITH SOURCES AND DATES OPENED

Item	Value	Source	Date opened
Server, Dell PowerEdge R760 (1 × Xeon Gold 6430, 256 GB)	$51,822.96 each	Dell online configurator (list price, default 36-month warranty)	Nov. 4, 2026
Disk, Dell 24 TB 7.2K NL-SAS (Seagate Exos X24)	$2,199 each	xByte, configurator list price	Nov. 4, 2026
Tape, LTO-9, 20-pack (HPE Q2079AH)	$2,190 per pack (list $3,234)	SHI, product listing	Nov. 4, 2026
Disk array, Dell PowerVault ME5084, xByte recertified, 2 SAS controllers	$19,999 each ($13,999 + 2 × $3,000)	xByte, configurator list price; 30-day warranty; 28-drive minimum	Nov. 4, 2026
Tape drive, LTO-9	$12,891.48 each	CDW, product listing	Nov. 1, 2026
Switch, Dell S4148T-ON	$7,424.99 each	Network Tigers, product listing (new)	Nov. 1, 2026
Inter-site link, 1 Gbps	$928.95 per month	Public procurement record, Govly, Apr. 2026 (proxy)	Nov. 1, 2026
Space rent	$195.94 per kW per month	CBRE Research, North America Data Center Trends H2 2025	Nov. 1, 2026
Electricity	13.37 cents per kWh	EIA, Electric Power Monthly, Table 5.3	Nov. 1, 2026
Server power	251 W idle, 748 W full load	SPEC, result power_ssj2008-20260406-01564	Nov. 1, 2026
Salary	$99,130 per year	BLS, Occupational Outlook Handbook	Oct. 29, 2026
Server life and PUE	41 months, 1.52	Uptime Institute, Global Data Center Survey 2026	Nov. 1, 2026
Broker capacity	5 million connections	EMQX Team, benchmark blog, Aug. 10, 2023 (vendor)	Nov. 1, 2026

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TABLE XXV
WORKED CALCULATION OF THE MONTHLY ON-PREMISES COSTS (USD)

Step	Calculation	Result
1. Server, per month	\$51,822.96 ÷ 41 months	1,263.97
2. Power, per server	499.5 W × 1.52 (PUE) × 730 h ÷ 1,000 × \$0.1337 per kWh	74.10
3. One server, per month	1,263.97 + 74.10 (unrounded 1,338.0772)	1,338.08
4. Brokers, application, database	2 ×, 1 × and 2 × the server cost	2,676.15; 1,338.08; 2,676.15
5. Disks and tape needed	Newest 90 days: 1,702 ÷ 12 × 3 = 425.5 TB per copy. Main site holds two copies: 851 TB ÷ 24 TB = 36 disks. Backup site holds one copy: 425.5 ÷ 24 = 18 disks, raised to the array minimum of 28. Older data: 1,276.5 TB per copy, × 3 = 3,829.5 TB, ÷ 18 TB = 213 cartridges = 11 packs of 20	64 disks; 213 cartridges
6. Storage purchase	64 × \$2,199 + 11 × \$2,190 + 2 × \$19,999 + 2 × \$12,891.48	230,606.96
7. Archive storage, per month	230,606.96 ÷ 41	5,624.56
8. Network and facilities, 1 site + backup	Switches: 2 × \$7,424.99 ÷ 41 = 362.19. Space: (5 × 0.4995 + 2 × 0.44) = 3.3775 kW × \$195.94 = 661.79	1,023.98
9. Network and facilities, 2 active sites	362.19 + (6 × 0.4995 + 2 × 0.44) kW × \$195.94 = 759.66, + link 928.95	2,050.80
10. Staff	\$99,130 ÷ 12 = 8,260.83 per engineer; × 4 on-premises, × 2 managed	33,043.33; 16,521.67
11. Break-even	Gap before staff ÷ cost per engineer: (34,547.10 − 13,338.93) ÷ 8,260.83; (34,547.10 − 15,703.83) ÷ 8,260.83	2.57 and 2.28 more engineers
12. Up-front purchases	5 servers + storage 230,606.96 + switches 14,849.98; 6 servers instead of 5 for two sites	504,571.74; 556,394.70

499.5 W is the midpoint of the idle (251 W) and full-load (748 W) power in the SPEC result, our assumption; 0.4995 kW is that figure per server and 0.44 kW is the listed rating of a switch. Totals in Table IX use unrounded values.

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